

Same Area, Different Perimeter

Mission

Warm-up and worked example

Name: _____

Date: _____

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YOUR MISSION Distinguish area from perimeter and explore their relationship.

SMART MOVE Name the attribute being measured before choosing a method.

Math Talk

Can two shapes have the same area but different perimeters?

My idea: _____

New Words

attribute	parallel	perpendicular
My meaning or picture:	My meaning or picture:	My meaning or picture:

Worked Example

Do this example with your teacher. Say what changes and what stays the same.

- 1. Build rectangles with area 24.
- 2. Calculate each perimeter.

What the example shows: _____

Same Area, Different Perimeter

Mission

Learn it and show your thinking

Name: _____

Date: _____

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YOUR MISSION Follow the learning path, then explain the big idea.

SMART MOVE Draw before asking for a rule.

[] **Step 1:** Build rectangles with area 24.

[] **Step 2:** Calculate each perimeter.

[] **Step 3:** Build rectangles with perimeter 20.

[] **Step 4:** Calculate each area.

[] **Step 5:** Discuss why area and perimeter measure different attributes.

My model, drawing, or organized work:

The big idea in my own words:

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Practice lab

Name: _____

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YOUR MISSION Use a model, equation, or explanation for every task.

SMART MOVE Check whether your answer is reasonable.

1. Name three defining attributes of a rectangle.

2. Find the area and perimeter of a 6-by-4 rectangle.

3. Draw two different rectangles with area 24 square units.

4. A square has perimeter 28 units. Find its side length and area.

5. Classify a quadrilateral with two pairs of parallel sides and four right angles.

6. Explain how two shapes can have the same area but different perimeters.

Choose one answer and prove it another way:

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Challenge and final check

Name: _____

Date: _____

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YOUR MISSION Try an unfamiliar problem and defend your reasoning.

SMART MOVE A wrong first idea can still lead to a strong solution.

Challenge Mission

Among rectangles with area 24, which has the smallest perimeter? Explain why.

My plan:

My solution and proof:

Final Check

Write one sentence using "square units" and one sentence using "units."

☐ I can teach it

☐ I need practice

☐ I need help

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Mission

Reusable math tool

Name: _____

Date: _____

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YOUR MISSION Use this page to draw, model, organize, and check.

SMART MOVE Name the attribute being measured before choosing a method.

Grid Lab

What does your drawing prove?
